

SECTION A [25 MARKS]

This section consists of 3 questions. Answer **all** questions.

1.

a) If $\lim_{x \rightarrow 1} \frac{5x^2 - px}{2x^2 - x} = 1$, find the value of p .

[3 marks]

b) Evaluate $\lim_{x \rightarrow +\infty} \left(\frac{8x^2 + 8x - 2}{x^2 + 1} \right)^{\frac{1}{3}}$.

[3 marks]

c) Find $\lim_{x \rightarrow 2^-} \frac{(3x + 2)|2 - x|}{2 - x}$.

[3 marks]

2. Find $\frac{dy}{dx}$ for each of the following.

a) $y = 3x^2 \ln(2x^3 - 5)^4$

[4 marks]

b) $y = \frac{1 + \cos x}{\sqrt{1 + \sin 3x}}$

[5 marks]

3. A ship is 6 km east and 8 km north of a lighthouse. It is moving north at a rate of 13 km/h and east at a rate of 16 km/h. At what rate is its distance from the lighthouse changing?

[7 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. If the modulus of the complex number $z = a + 4\sqrt{2} + (a - 4\sqrt{2})i$ is $2\sqrt{21}$, where $a > 0$, find the values of a . Hence, express z in polar form.
[5 marks]
2. Solve
 - a) $1 + \sqrt{x+3} = \sqrt{2x-1}$
[5 marks]
 - b) $\left| \frac{x+2}{x-4} \right| < 5$
[6 marks]
3. It is given that $f(x) = 2 - \ln|3-x|$.
 - a) State domain and range of $f(x)$.
[2 marks]
 - b) Find $f^{-1}(x)$ if it exists.
[4 marks]
4.
 - a) Let f and g be two functions. Given that $g \circ f(x) = \frac{x+2}{3}$ and $g(x) = 2x+5$, find $f(x-1)$.
[5 marks]
 - b) Given that $h(x) = \begin{cases} 4e^x, & x < 0 \\ \frac{3}{4}x + 4, & 0 \leq x < 4 \\ x^2 - 8x + 18, & x \geq 4 \end{cases}$.
 - i. Sketch $h(x)$.
[5 marks]
 - ii. Find the domain and range for $h(x)$ and $h^{-1}(x)$.
[4 marks]

5.

- a) Find the vertical and horizontal asymptotes of $f(x) = \frac{x+4}{\sqrt{x^2-25}}$.

[9 marks]

- b) Given the function $h(x) = \begin{cases} e^x + 2, & x < 0 \\ x^2 + 3, & x \geq 0 \end{cases}$. Determine the continuity of h at $x = 0$.

[5 marks]

6.

- a) Given $y = \frac{x}{2x+1}$, find;

i. $\frac{dx}{dy}$

[4 marks]

- ii. $\frac{d^2x}{dy^2}$ in terms of y and x .

[4 marks]

- b) Given the parametric equations for the curve $x = \frac{2}{\sqrt{2-t^2}}$ and $y = \frac{\sqrt{2-t^2}}{t}$. Find

$\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ when $t = 1$.

[8 marks]

7. A circle and a square is to be formed by using a wire with total length 8 cm. The radius of the circle x cm while one of the sides of the square is y cm.

- a) By considering the perimeter of both shapes, find an expression for y in terms of x .

[2 marks]

- b) Show that the total area of the two shapes is given by:

$$A = \left(\frac{\pi^2 + 4\pi}{4} \right) x^2 - 2\pi x + 4$$

[3 marks]

- c) If the total area of the two shapes is the minimum possible, what percentage of the wire is used for the circle?

[4 marks]

SUGGESTED ANSWERS

SECTION A

1. a) $p=4$ b) 2 c) 8
2. a) $6x \ln(2x^3 - 5)^4 + \frac{72x^2}{(2x^3 - 5)}$ b) $\frac{-2 \sin x(1 + \sin 3x) - 3 \cos 3x(1 + \cos x)}{2(1 + \sin 3x)^{\frac{3}{2}}}$
3. 20 km/h

SECTION B

1. $a = \sqrt{10}$, $z = 2\sqrt{21}(\cos(-0.276) + i \sin(-0.276))$
2. a) $x=13$ b) $\therefore \{x : x < 3 \text{ or } x > \frac{11}{2}\}$
3. a) $D_f = (-\infty, 3)$ $R_f = \mathbb{R}$ b) $f^{-1}(x) = 3 - e^{2-x}$
4. a) $f(x-1) = \frac{x-14}{6}$ b) ii. $D_h = R_{h^{-1}} = (-\infty, \infty)$
 $R_h = D_{h^{-1}} = (0, \infty)$
5. a) $x=5, x=-5, y=1, y=-1$ b) continuous
6. a) i. $\frac{dx}{dy} = \frac{2x+1}{1-2y}$ ii. $\frac{8x-8y-16xy+4}{(1-2y)^3}$
b) -2 and $5/2$
7. a) $y = 2 - \frac{\pi x}{2}$ c) 43.99%